

MS233 - Advancing Computational Mechanics with Equation Discovery and Symbolic Regression

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Symbolic regression (SR) is rooted in machine learning and symbolic computation and enables finding potentially interpretable equations directly from observational datasets.

Guiding Genetic Programming with Description Length Improves Symbolic Regression Solutions

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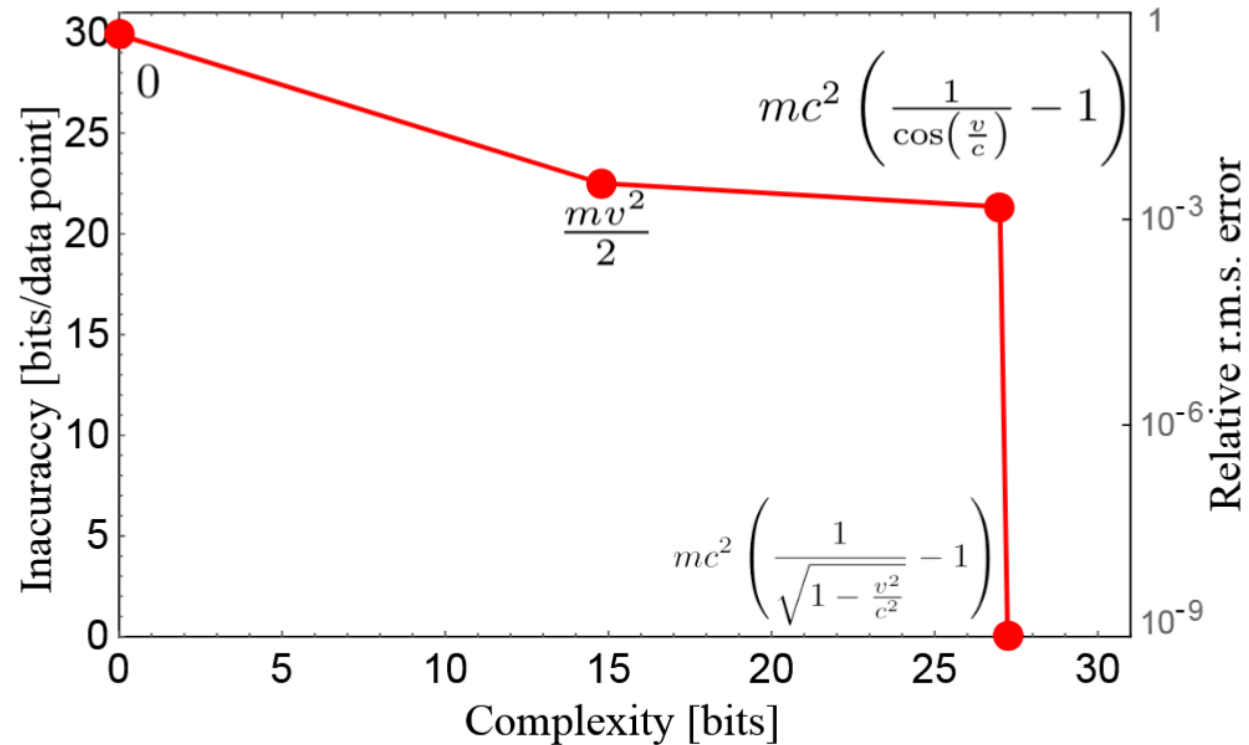
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Figure from Udrescu et al. 2020, *AI Feynman 2.0: Pareto-optimal symbolic regression ...*



The **Pareto front** contains models for kinetic energy with different tradeoffs between different accuracy and complexity.

Genetic programming for symbolic regression (GPSR) is a cyclic process of **model creation** and **model selection**.

Their interaction drives GP search.

Typical approach in two steps:

1. Search for candidates with minimal training loss.
2. Select a model from the final candidate set based on various heuristics.

Aim

Can we "improve" GP search (1) and model selection (2) by minimizing description length (MDL)?

Improve:

- produce solutions with smaller (or equal) test error
- and shorter expressions

Information theory: study of communication of information through a channel.

The **description length (DL)** is the total length of the message required to communicate a dataset.

$$DL = L(D|M) + L(M)$$

Application of Occham's razor.

Prior work on using MDL for symbolic regression:

- STROGANOFF (Iba et al, 1993)
- AI Feynman 2.0 (Udrescu et al., 2020)
- Exhaustive Symbolic Regression (**Bartlett et al. 2023**)
- MDLFormer (Yu et al., 2025)

Description length (Bartlett et al. 2023):

$$\begin{aligned} & -\log(\mathcal{L}(\hat{\boldsymbol{\theta}})) + k \log(n) + p \log(2) + \sum_j \log(c_j) \\ & + \sum_i^p \log(|\hat{\theta}_i|/\Delta_i), \end{aligned}$$

The $k \log n$ term penalizes structural complexity k with n distinct symbols

Only the quantization grid Δ_i for the MLE $(\hat{\theta})$ is a free parameter.

It turns out optimal Δ_i can be calculated from Fisher information.

| D. Bartlett, H. Desmond, P.G. Ferreira: Exhaustive Symbolic Regression, 2023

We tested three variants of using DL in GPSR

- Multi-objective minimization of training loss and length and **select a model from the Pareto front** using one of the criteria (MO-Length).
- Multi-objective minimization of training loss **and model complexity** ($L(M)$). Select the MDL model from the Pareto front. (MO-DL)
- Single-objective **minimization of DL directly**. (SO-DL)

Experiments:

- 34 Datasets x 100 runs; synthetic and real-world datasets
- For synthetic datasets we know the generating function and the irreducible noise σ_{err} , but not given to GP.

We compare several model selection criteria for MO-Length:

$$\text{AIC} = -2 \log \mathcal{L} + 2p$$

$$\text{BIC} = -2 \log \mathcal{L} + p \log m$$

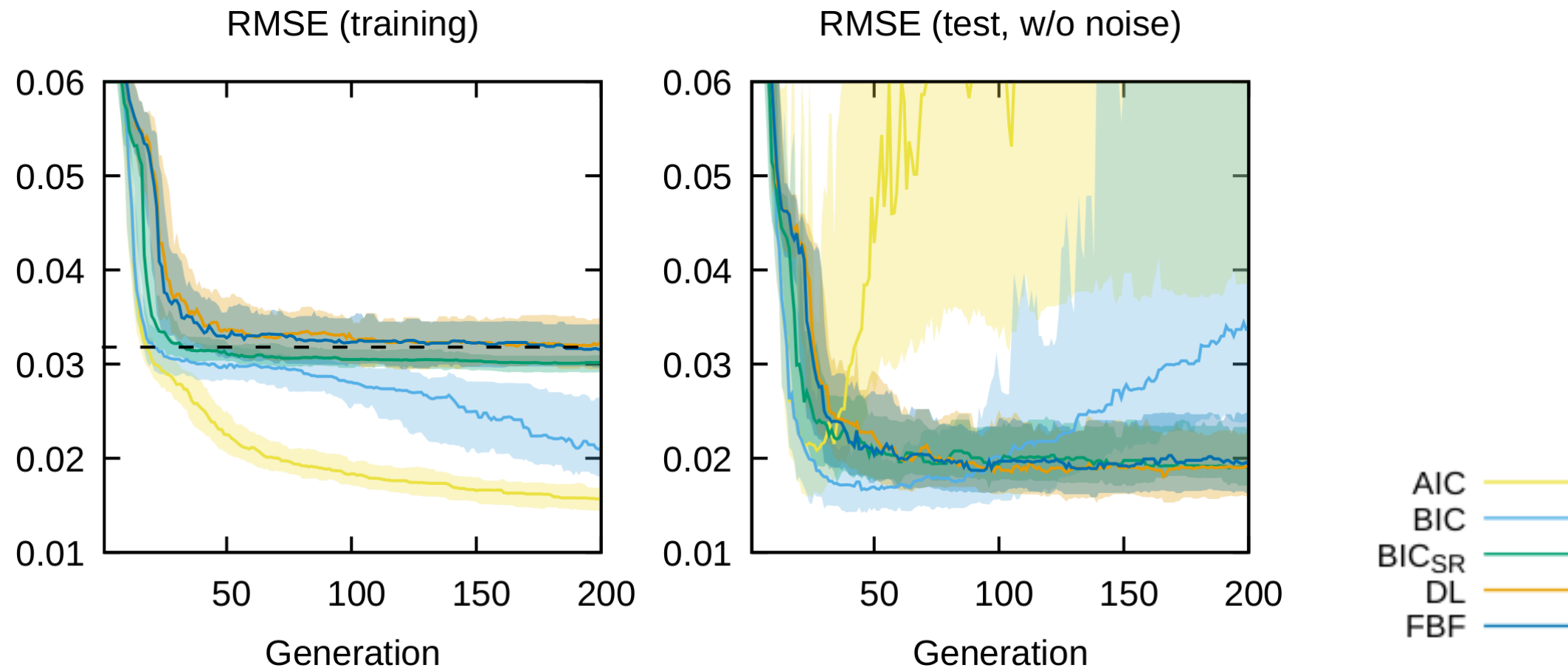
$$\text{BIC}_{\text{SR}} = -2 \log \mathcal{L} + 2k \log n + p \log m$$

$$\text{FBF} = -(1 - b) \log \mathcal{L} + k \log n + p \log m + \frac{p}{2} (\log(2\pi e^{1-\log 3})), \quad b = m^{-1/2}$$

$$\text{DL} = \log \mathcal{L} + k \log n + \sum_i^p \log |\hat{\theta}| / \Delta_i$$

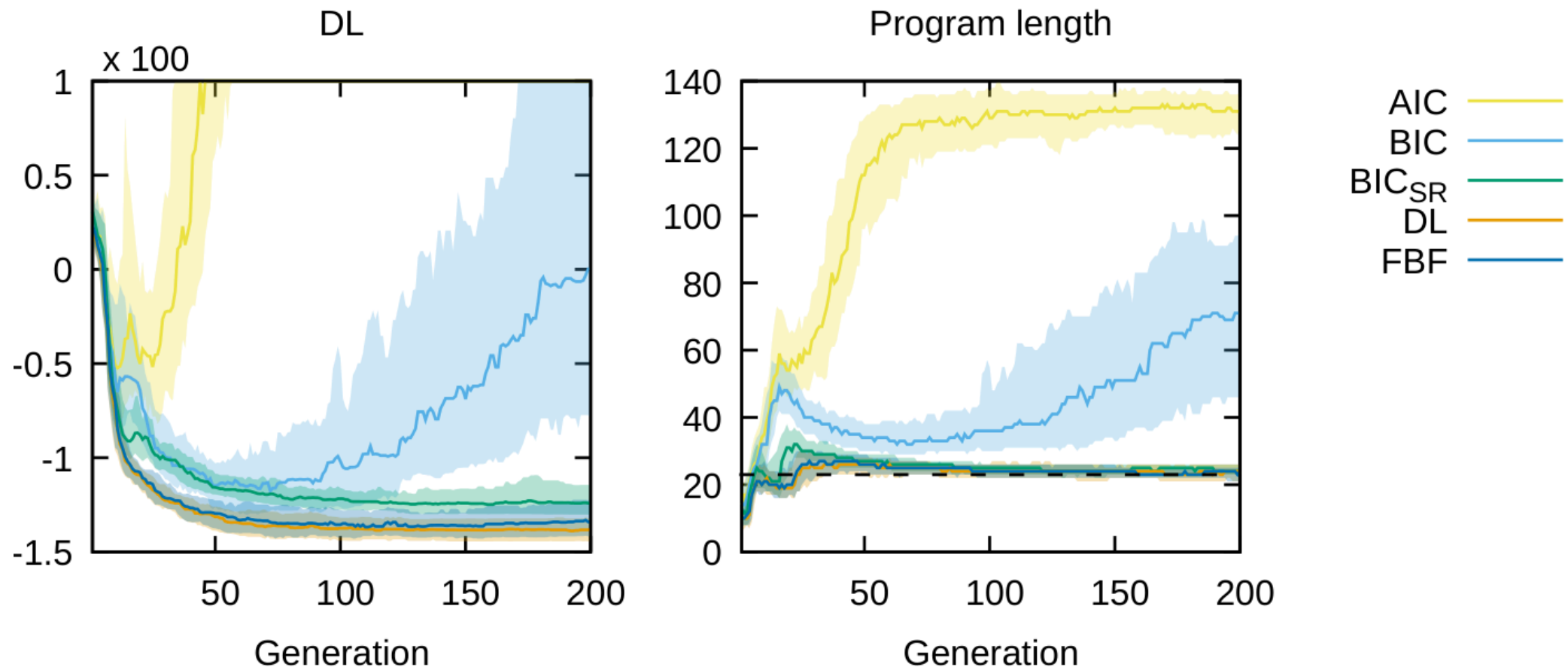
BIC_{SR} , FBF , and DL are different approximations to Bayesian model evidence and all include the structural complexity penalty.

Exemplary **MO-Length** result on a synthetic dataset



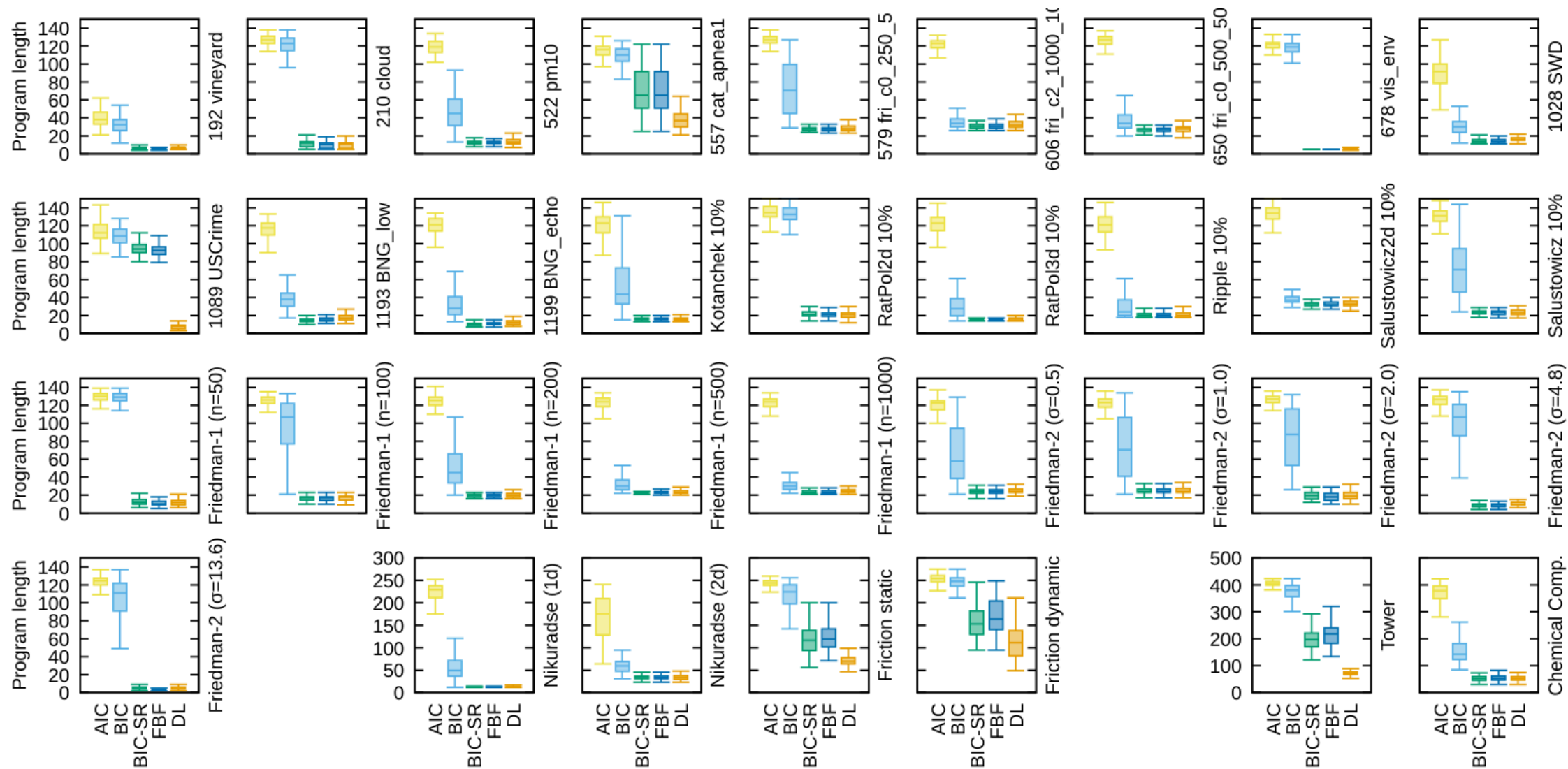
FBF, DL approach irreducible noise level

AIC and BIC overfit. BIC_{SR}, FBF, DL are similar. BIC_{SR} overfits slightly.

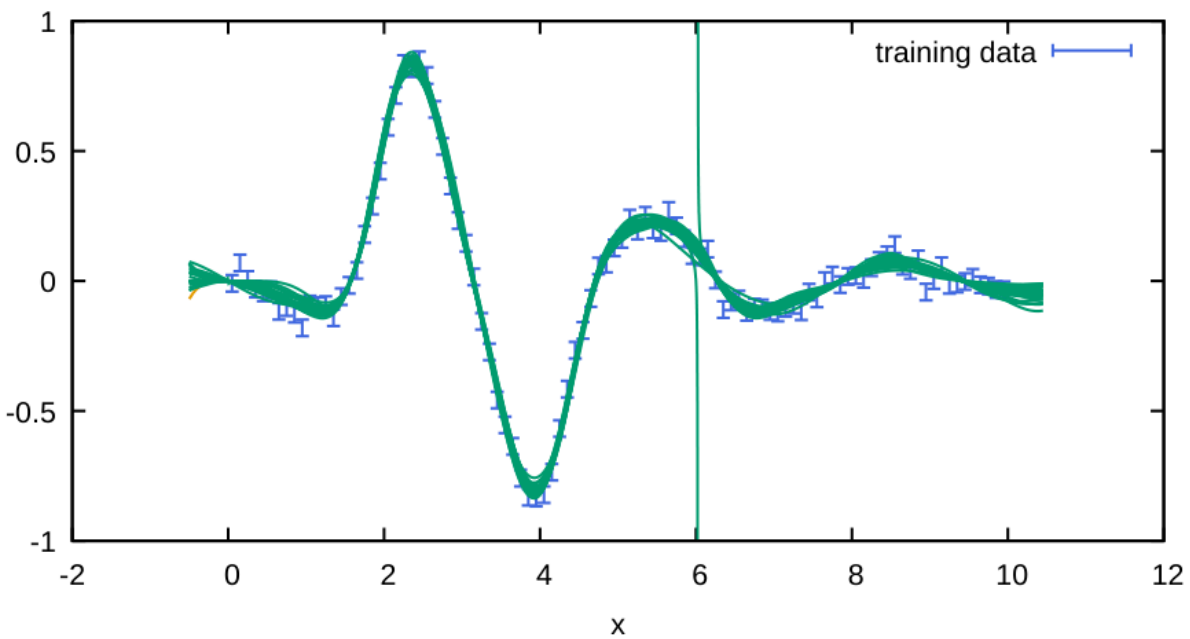


BIC_SR, DL, FBF all converge to length of generating expression

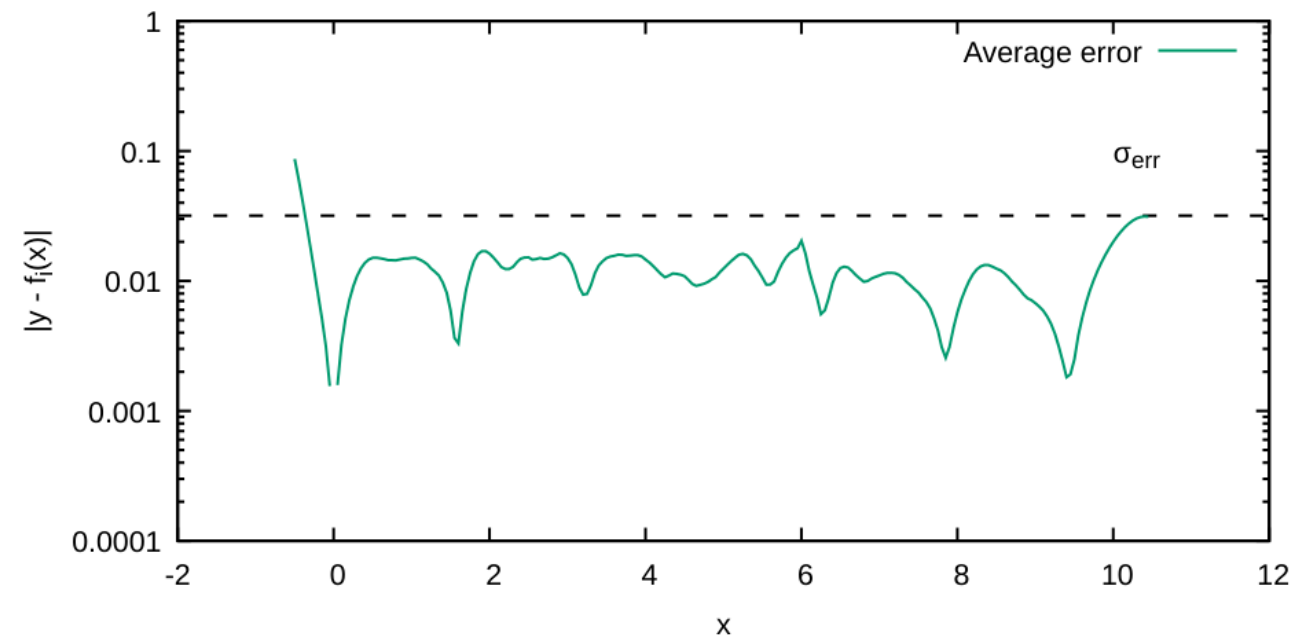
DL selects shortest models



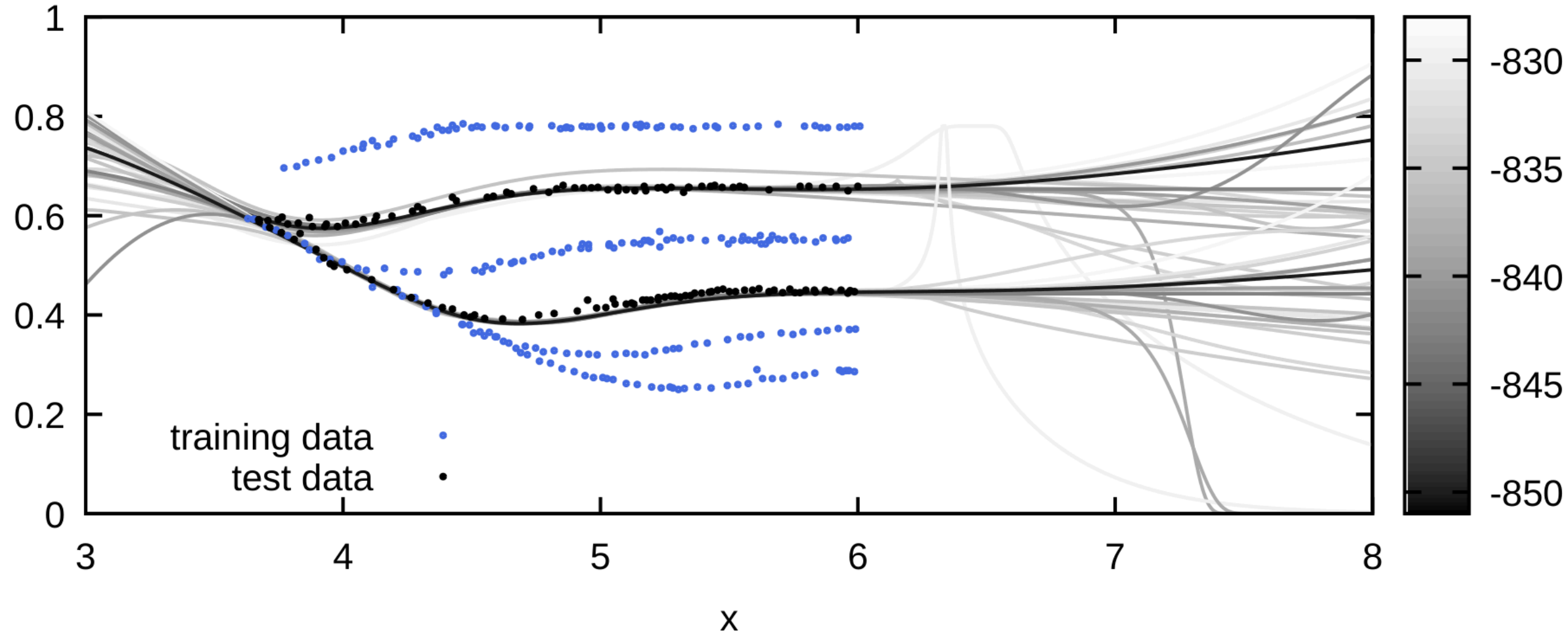
Prediction of MDL solutions found with MO-Length (Salustowicz, 10% noise)



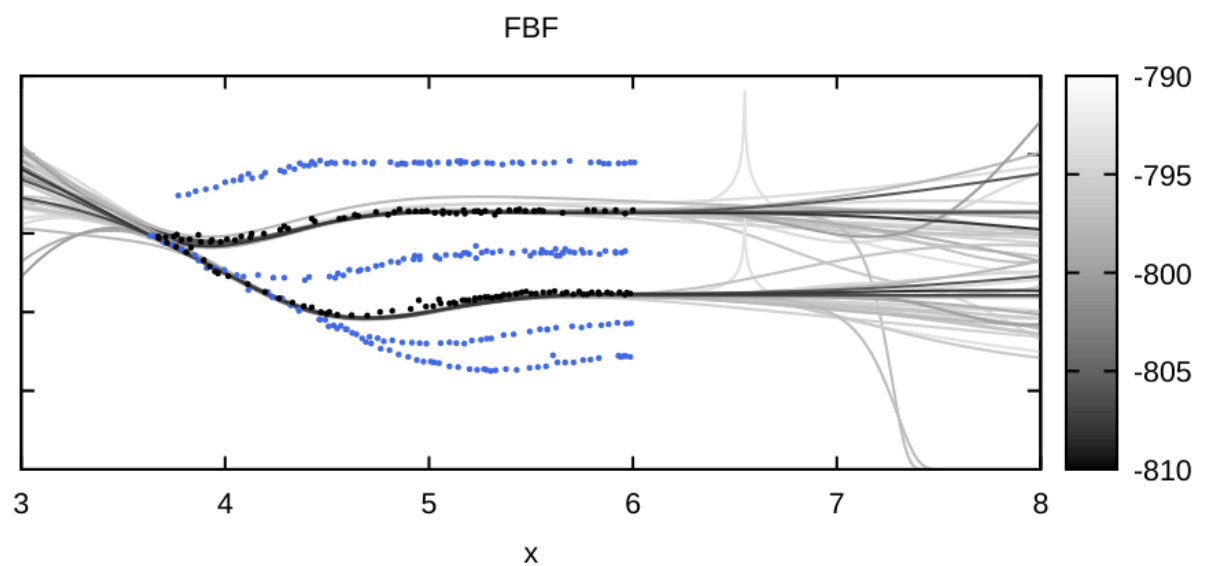
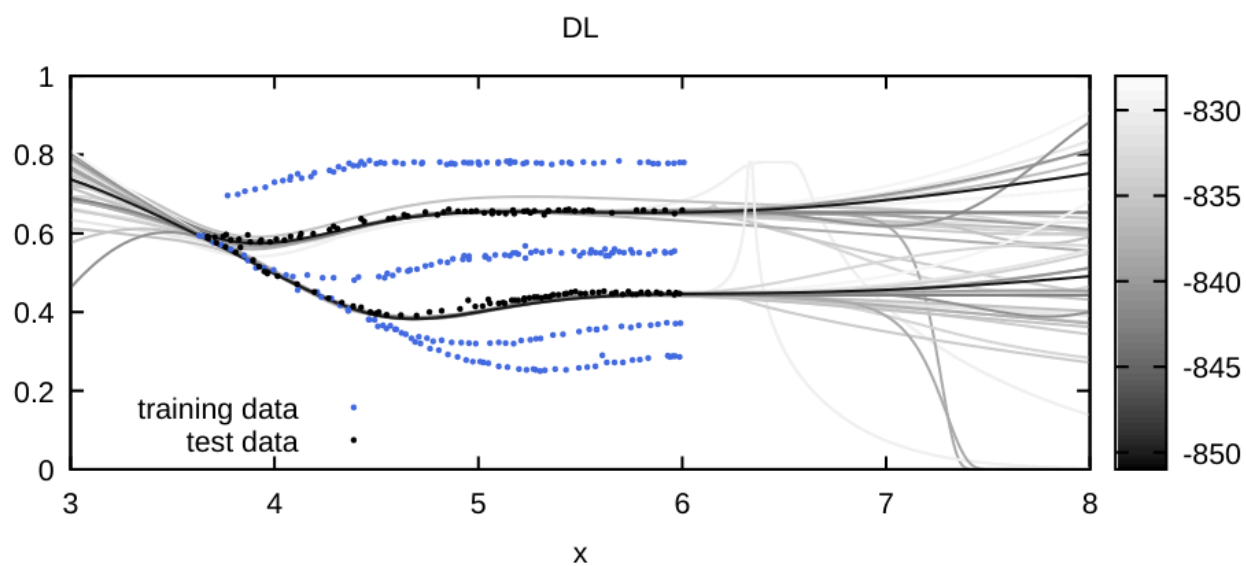
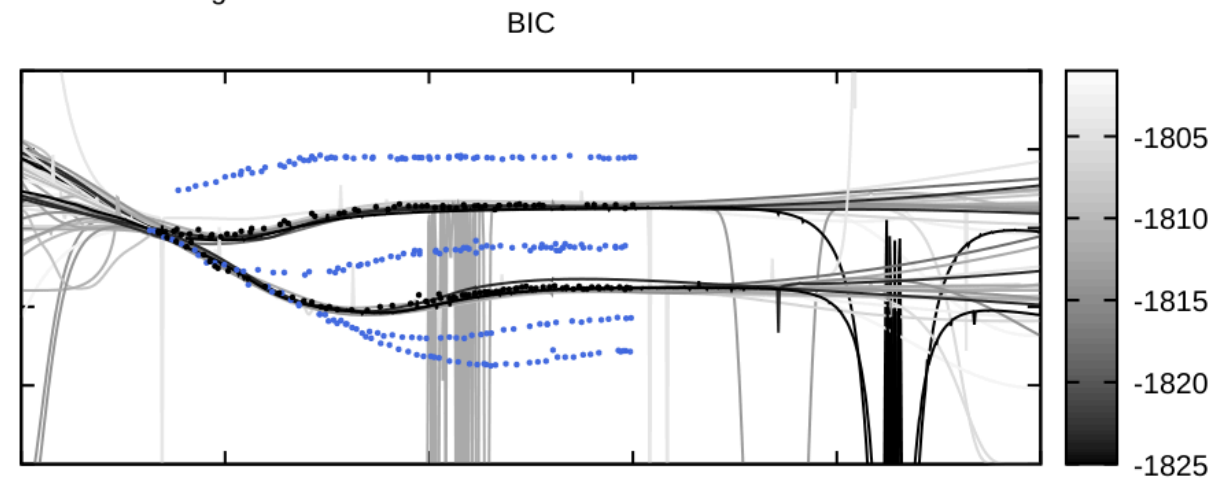
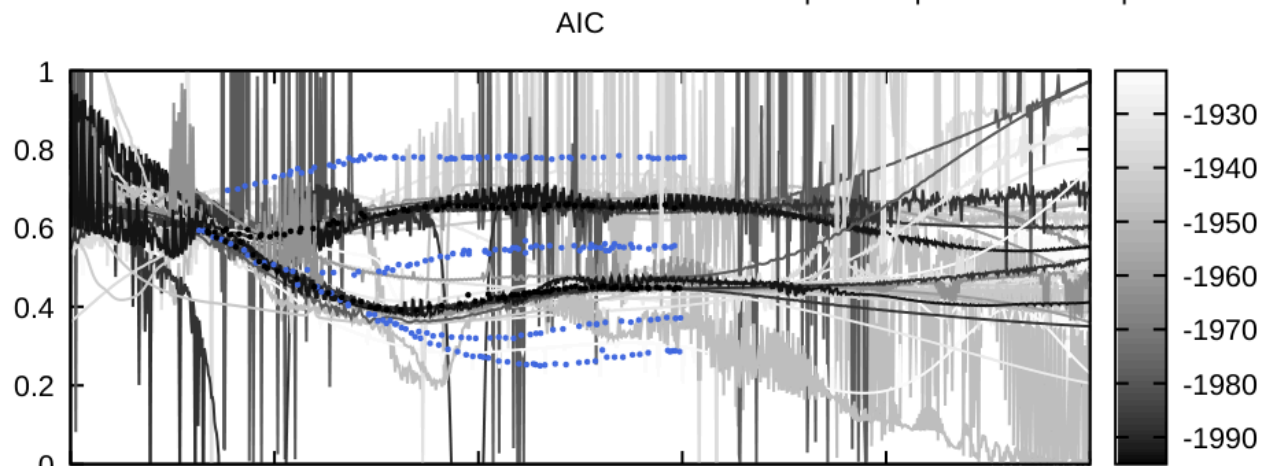
Average absolute error of solutions



DL

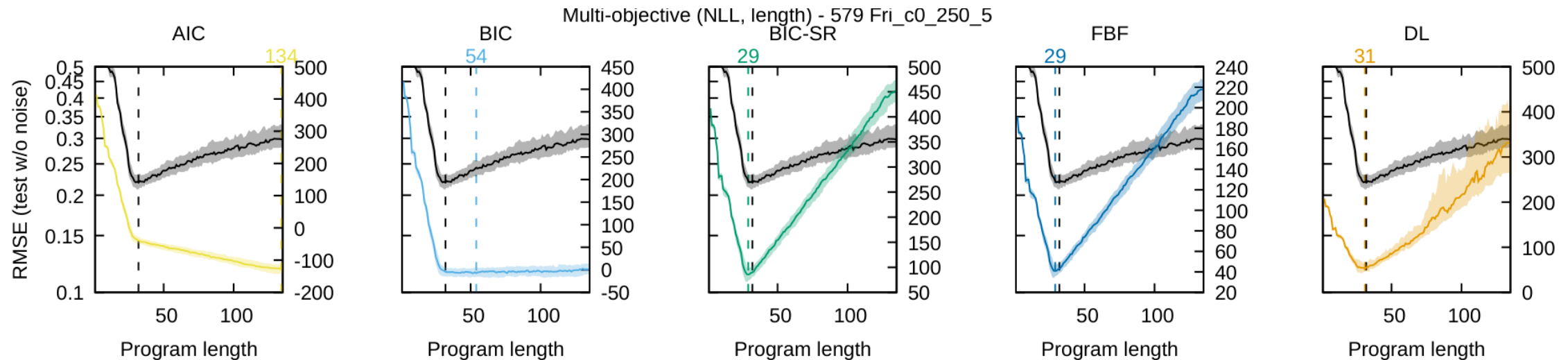


Extrapolation predictions of top-25 solutions found with MO-Length for Nikuradse-1



Detailed analysis of the final Pareto fronts shows a **clear minimum close to minimum test error** for BIC_SR, FBF, and DL.

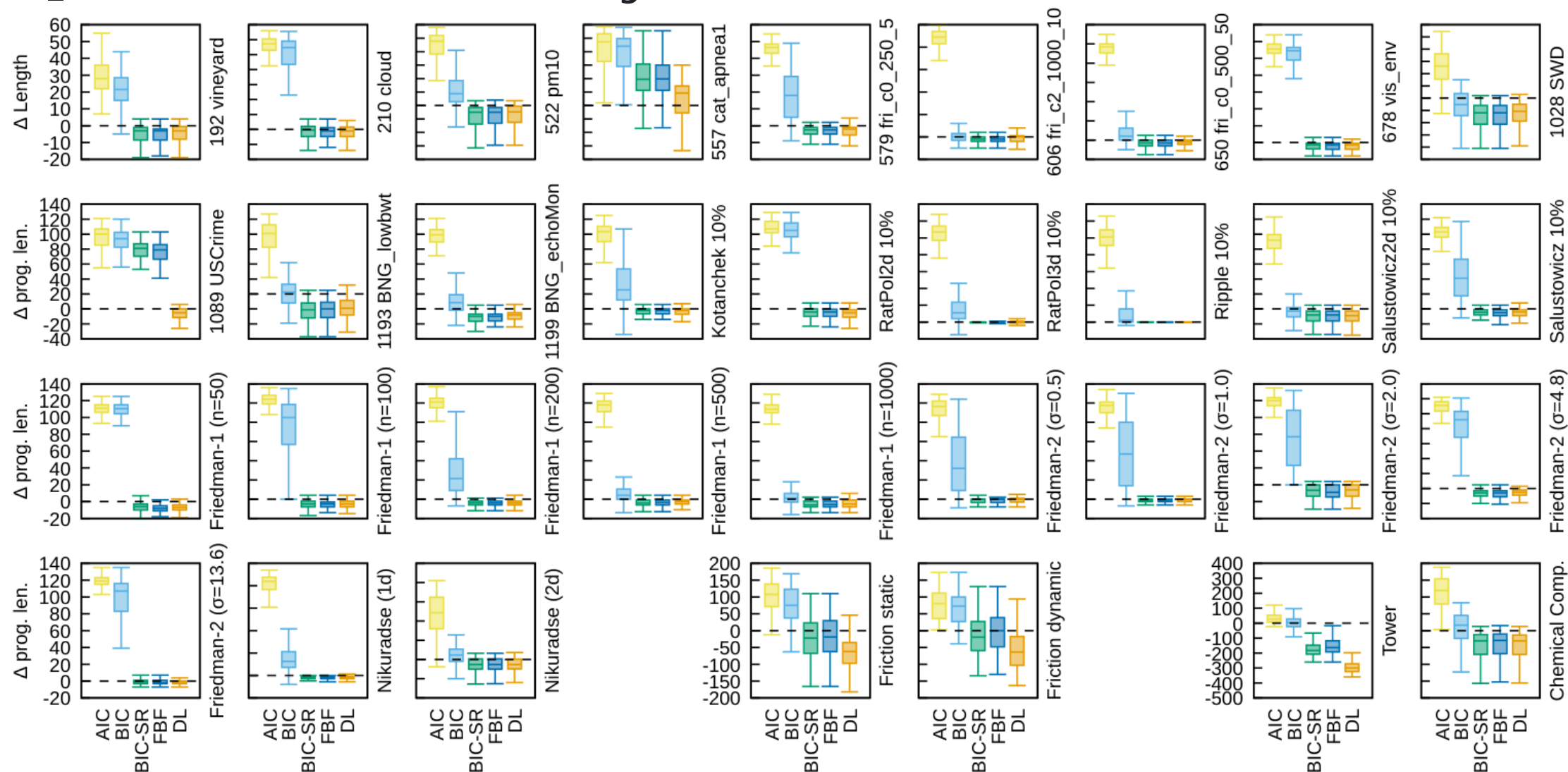
AIC selects overfit solutions, BIC is flat.



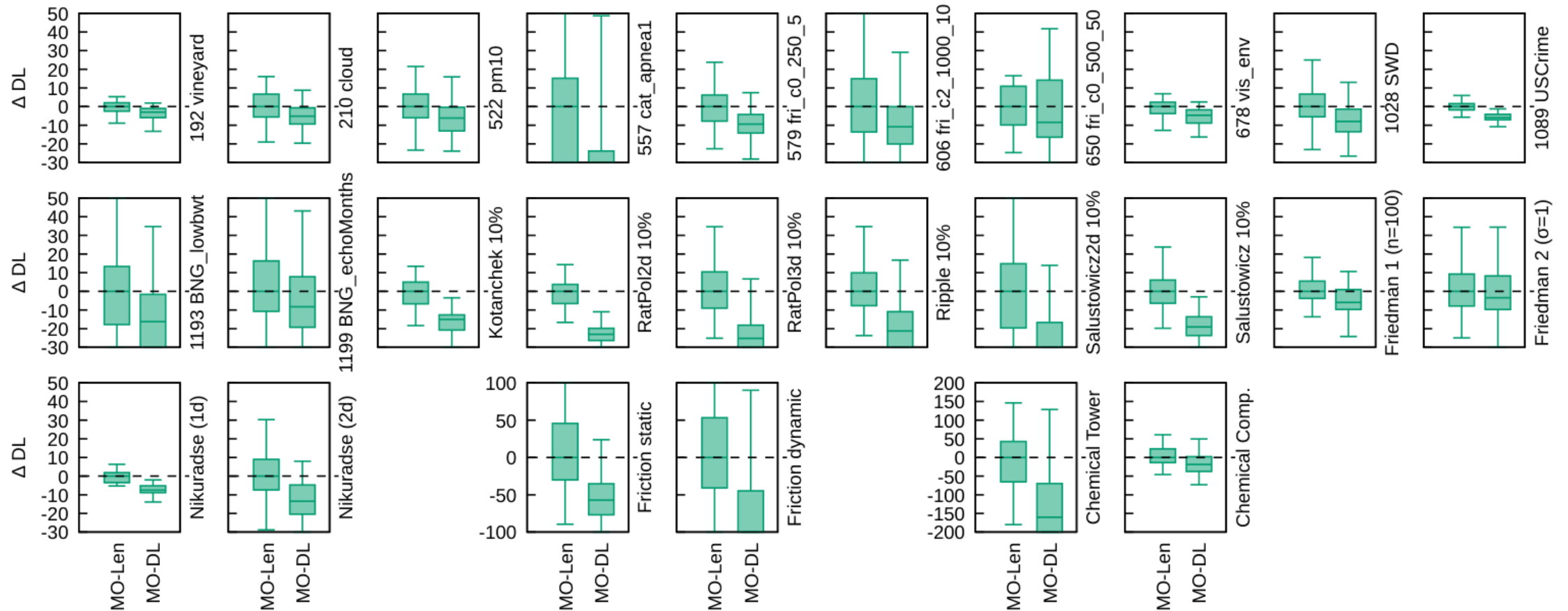
Colors: AIC, BIC, BIC_SR, FBF, DL

Black: test error of solution candidates in the Pareto front.

BIC_SR, FBF, DL select models with length close to the best test error models.



Using model complexity instead of length as the second objective (MO-DL vs. MO-Length) produces expressions with better DL (but no difference in test errors).



Summary and Conclusions

- **BIC_SR, FBF, DL all work well** for GPSR model selection.
- For large datasets they produce similar results, for small datasets DL slightly better.
- **BIC_SR and FBF cost almost nothing**, DL calculation requires FIM.
- MO-DL produces smaller DL than MO-Length but no relevant difference in test errors.
- Single-objective optimization for BIC, BIC_SR, FBF, DL fails because of early loss of diversity

Limitations

- Only Gaussian likelihoods and limited set of datasets
- Simple function complexity penalty ($k \log n$)
- No rigorous comparison to PySR or pyoperon

Questions



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Parameter	Value
Population size	1000
Selection	Tournament (group size 2)
Max. generations	200 (Synthetic, SRbench, Nikuradse) 400 (Friction, Tower)
Crossover prob.	90%
Mutation prob.	10% of indiv. and 5% for each node
Initialization	Grow with depth limit 5
Max. length	100 (Synthetic, SRbench) 200 (Nikuradse, Friction) 300 (Tower, Chem. comp.)
Function set	$\{+, -, \times, \div, x^2, \sin, \cos, \exp, \log x , \sqrt{ x }, x ^y\}$
Terminal set	$\{\text{var}, \text{param}, \text{var} \times \text{param}\}$

Some background on Bayesian model selection:

Posterior probability for a model f_i :

$$\begin{aligned} P(f_i|D) &= \frac{1}{P(D)} \int P(D|f_i, \theta_i) P(\theta_i|f_i) P(f_i) d\theta_i \\ &\equiv \frac{P(f_i)}{P(D)} \mathcal{Z}(D|f_i) \end{aligned}$$

For a fixed dataset we may ignore (constant) $P(D)$:

$$-\log P(f_i|D) = -\log P(f_i) - \log \mathcal{Z}(D|f_i)$$

The Bayesian evidence $\mathcal{Z}$ is difficult to evaluate and is typically approximated.

The Laplace approximation around the maximum likelihood estimate (MLE) can be used:

$$\begin{aligned}\log \mathcal{Z}(D|f_i) &\approx \log P(D|f_i, \hat{\theta}_i) + \log P(\hat{\theta}_i|f_i) \\ &\quad + \frac{p}{2} \log(2\pi) - \frac{1}{2} \log |\text{FIM}(\hat{\theta}_i)|\end{aligned}$$

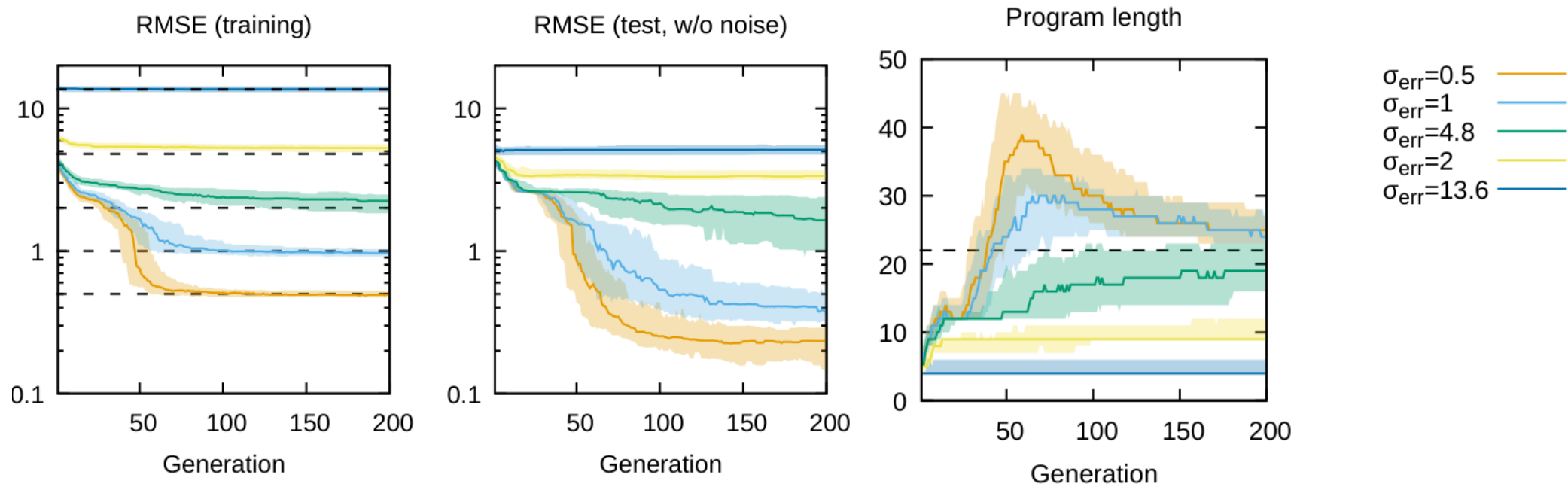
- $|\text{FIM}|$ is the Fisher information matrix evaluated at the MLE.
- As $n \rightarrow \infty$ the prior and the $p/2 \log(2\pi)$ terms can be ignored and the last term approaches $p/2 \log n$. The result is 1/2 BIC

Details on the calculation of total DL

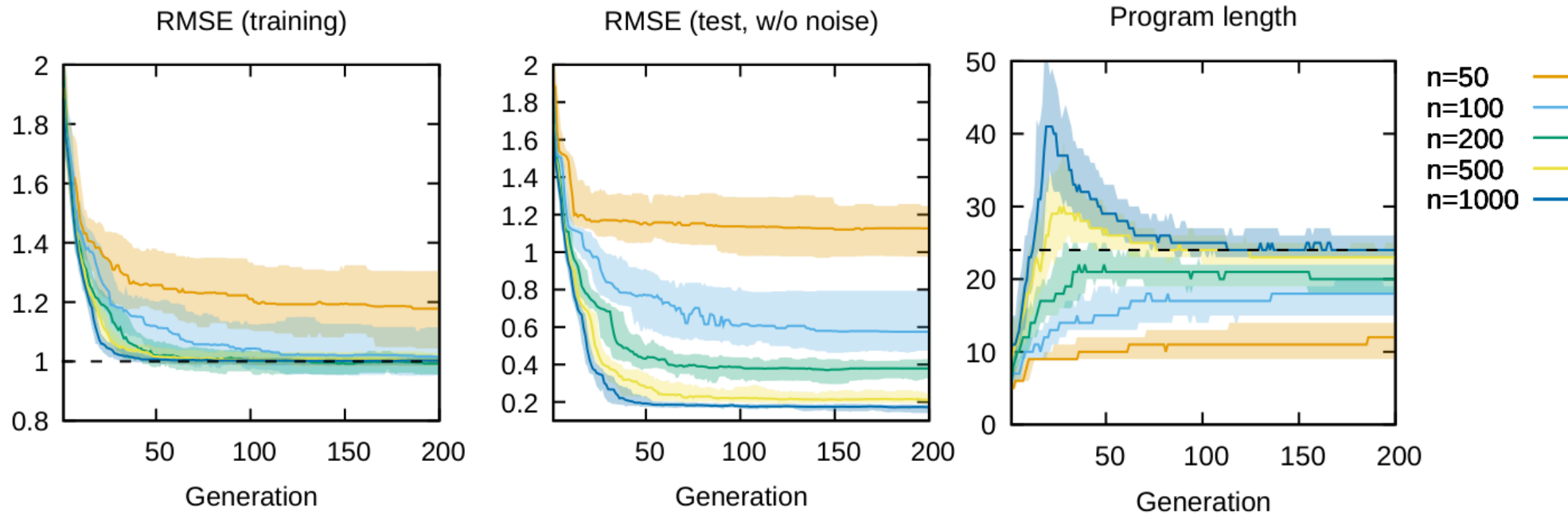
$$\begin{aligned}
 \text{DL} = & -\log \mathcal{L}(x, y | \hat{\theta}, \hat{\sigma}_{\text{err}}^2) + \underbrace{k \log n + \sum_{c_i} \log c_i}_{\text{function complexity}} \\
 & + \underbrace{\frac{1}{2} \sum_{i=1}^p \left(\max(0, \log S_{ii} - \log 3 + \log |\hat{\theta}_i^{\text{rot}}|) \right)}_{\text{parameter complexity}}
 \end{aligned}$$

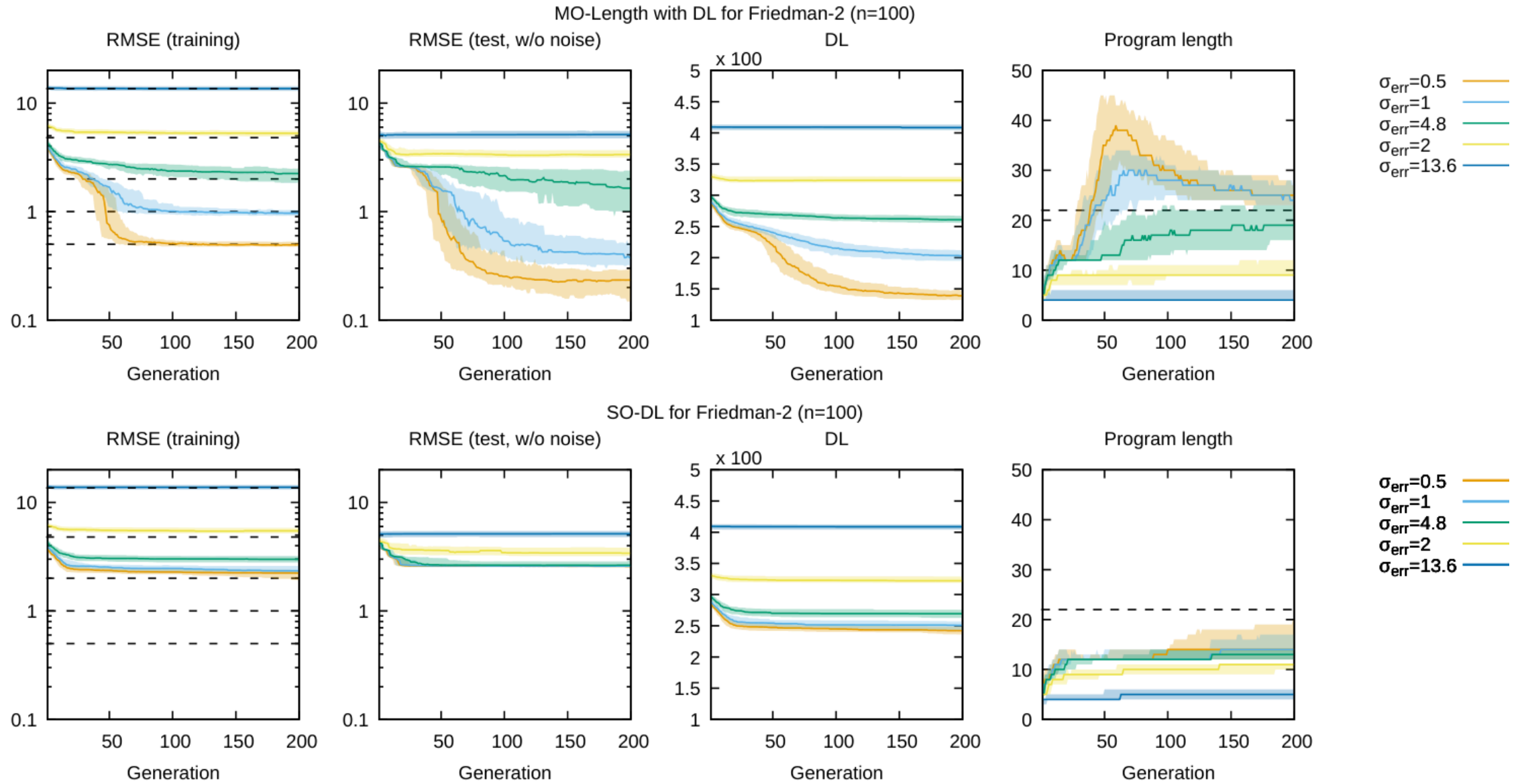
Parameter complexity: $\text{FIM} = USV^T$, $\log S_{ii}$, $\hat{\theta}_{\text{rot}} = V\hat{\theta}$

Effect of noise level



Effect of training set size





Datasets

Identifier	Observations	Vars.
Synthetic		
Friedman-1 (additive)	(50 – 1000, 1000)	10
Friedman-2 (interactions)	(100, 1000)	10
Kotanchek	(100, 2025)	3
RatPol2d	(50, 1156)	2
RatPol3d	(300, 2700)	3
Ripple	(300, 1000)	2
Salustowicz	(100, 220)	1
Salustowicz2d	(606, 2553)	2
Engineering		
Nikuradse (1 dim)	(230, 132)	1
Nikuradse (2 dim)	(201, 163)	2
Friction dyn.	(1009, 1007)	17
Friction stat.	(1009, 1007)	16
Chemical - Tower	(3136, 1865)	25
Chemical - Competition	(711, 355)	57
SRBench selection		
192 Vineyard	(31, 21)	2
210 Cloud	(64, 44)	5
522 PM10	(300, 200)	7
557 Anacat Apneal	(285, 190)	3
579 Fri. c0 250 5	(150, 100)	5
606 Fri. c2 1000 10	(600, 400)	10
650 Fri. c0 500 50	(300, 200)	50
678 Vis. env	(66, 45)	3
1028 SWD	(600, 400)	10
1089 USCrime	(28, 19)	13
1193 BNG lowbwt	(1000, 10368)	9
1199 BNG echo months	(1000, 5832)	9

Dataset	AIC	BIC	BIC _{SR}	FBF	DL
Kotanchek	—	0.9919	0.9965	0.9960	0.9952
RatPol2d	−0.9268	−0.4004	0.9719	0.9745	0.9680
RatPol3d	0.9922	0.9991	0.9998	0.9998	0.9998
Ripple	0.9963	0.9996	0.9998	0.9998	0.9998
Salustowicz	0.9793	0.9904	0.9960	0.9960	0.9960
Salustowicz2d	0.9957	0.9985	0.9982	0.9983	0.9983
Friedman-1 $n = 50$	−1.8290	−1.6330	0.8247	0.7934	0.7964
Friedman-1 $n = 100$	0.5202	0.7211	0.9467	0.9433	0.9470
Friedman-1 $n = 200$	0.8700	0.9494	0.9760	0.9760	0.9772
Friedman-1 $n = 500$	0.9463	0.9898	0.9923	0.9923	0.9929
Friedman-1 $n = 1000$	0.9760	0.9958	0.9952	0.9951	0.9954
Friedman-2 $\sigma = 0.5$	0.9799	0.9898	0.9981	0.9981	0.9979
Friedman-2 $\sigma = 1.0$	0.9185	0.9662	0.9952	0.9952	0.9945
Friedman-2 $\sigma = 2.0$	0.5151	0.7375	0.9051	0.8863	0.8990
Friedman-2 $\sigma = 4.8$	−1.7300	−1.6330	0.4936	0.4804	0.5709
Friedman-2 $\sigma = 13.6$	−39.4100	−19.4500	−0.1464	−0.1252	− 0.0154
Nikuradse-1 (2d)	0.9456	0.9945	0.9947	0.9947	0.9947
Nikuradse-2 (1d)	0.9467	0.9741	0.9820	0.9820	0.9820
Chem. 1 Tower	0.9582	0.9579	0.9518	0.9532	0.9346
Chem. 2 Comp.	0.1749	0.6681	0.7072	0.7164	0.7072
Friction dyn.	0.9143	0.9226	0.9428	0.9421	0.9341
Friction stat.	0.9096	0.9124	0.9161	0.9167	0.9138
192 vineyard	0.2191	0.2688	0.5562	0.5512	0.5729
210 cloud	−1.9630	−1.1920	0.7910	0.7972	0.7959
522 pm10	0.0375	0.2010	0.2448	0.2448	0.2267
557 analcat apnea1	0.4781	0.5416	0.6411	0.6382	0.8007
579 fri c0 250 5	0.9087	0.9350	0.9514	0.9515	0.9486
606 fri c2 1000 10	0.9741	0.9784	0.9784	0.9784	0.9784
650 fri c0 500 50	0.9341	0.9549	0.9566	0.9566	0.9566
678 vis. env.	−1.6130	−1.3320	0.2915	0.3016	0.2833
1028 SWD	0.3638	0.3752	0.3629	0.3639	0.3650
1089 USCrime	−0.0618	−0.1625	−0.1394	−0.1148	0.7147
1193 BNG lowbwt	0.5405	0.5748	0.5674	0.5681	0.5718
1199 BNG echoMon	0.1728	0.4229	0.4245	0.4249	0.4277
Win prob. (34 × 100 pairings)	DL>AIC 88.8% [†]	DL>BIC 73.0% [†]	DL>BIC _{SR} 51.8%	DL>FBF 51.7%	

Dataset	SO-AIC	SO-BIC	SO-BIC _{SR}	SO-FBF	SO-DL	MO-Len DL
Kotanchek	—	0.9924	0.9881	0.9824	0.9858	0.9952
RatPol2d	−0.1896	0.3601	−0.1423	−0.1292	−1.4090	0.9680
RatPol3d	0.9991	0.9997	0.9996	0.9996	0.9996	0.9998
Ripple	0.9907	0.9984	0.8857	0.8858	0.8873	0.9998
Salustowicz	0.9880	0.9965	0.9784	0.9795	0.9817	0.9960
Salustowicz2d	0.9976	0.9965	0.9922	0.5953	0.8783	0.9983
Fried.-1 $n = 100$	0.7143	0.9481	0.8576	0.8516	0.8524	0.9470
Fried.-1 $n = 50$	−0.8257	0.8272	0.7700	0.7602	0.7802	0.7964
Fried.-1 $n = 200$	0.9497	0.9794	0.9474	0.9250	0.9251	0.9772
Fried.-1 $n = 500$	0.9879	0.9906	0.9785	0.9798	0.9795	0.9929
Fried.-1 $n = 1000$	0.9941	0.9946	0.9896	0.9899	0.9918	0.9954
Fried.-2 $\sigma = 0.5$	0.9920	0.9635	0.7480	0.7446	0.7432	0.9979
Fried.-2 $\sigma = 1.0$	0.9712	0.9088	0.7414	0.7401	0.7420	0.9945
Fried.-2 $\sigma = 2.0$	0.7618	0.8730	0.7227	0.7236	0.7326	0.8990
Fried.-2 $\sigma = 4.8$	−0.5562	0.5923	0.4691	0.4630	0.5535	0.5709
Fried.-2 $\sigma = 13.6$	−7.8540	−0.3393	−0.0481	−0.0695	−0.0186	−0.0154
Nikuradse 1 (2d)	0.8521	0.8460	0.9883	0.9851	0.9787	0.9947
Nikuradse 2 (1d)	0.9795	0.9819	0.9818	0.9820	0.9819	0.9820
Chem. 1 Tower	0.9486	0.9474	0.9322	0.9339	0.9218	0.9346
Chem. 2 Comp.	0.5186	0.6522	0.5897	0.5472	0.4806	0.7072
Friction dyn.	0.9177	0.9179	0.8648	0.8696	0.8659	0.9341
Friction stat.	0.9139	0.9078	0.8528	0.8587	0.8175	0.9138
192 vineyard	0.4276	0.5710	0.5857	0.5805	0.5889	0.5729
210 cloud	−2.4730	0.7212	0.8160	0.8089	0.7990	0.7959
522 pm10	0.1690	0.2014	0.1365	0.1410	0.1551	0.2267
557 analcat apneal	0.7320	0.7821	0.8677	0.8842	0.8577	0.8007
579 fri c0 250 5	0.9294	0.9472	0.7633	0.7559	0.8156	0.9486
606 fri c2 1000 10	0.9762	0.9778	0.9765	0.9770	0.9771	0.9784
650 fri c0 500 50	0.9023	0.8690	0.8228	0.8376	0.8594	0.9566
678 vis. env.	−0.0044	0.2711	0.3016	0.3137	0.2947	0.2833
1028 SWD	0.3788	0.3715	0.3451	0.3485	0.3718	0.3650
1089 USCrime	−0.9404	0.6968	0.7307	0.7351	0.7207	0.7147
1193 BNG lowbwt	0.5750	0.5738	0.5612	0.5647	0.5676	0.5718
1199 BNG echoMon	0.4173	0.4341	0.4254	0.4256	0.4275	0.4277
Win prob. (3400 pairings)	DL>AIC 79.5 % [†]	DL>BIC 61.8 % [†]	DL>BIC _{SR} 78.3 % [†]	DL>FBF 77.4 % [†]	DL>SO-DL 76 % [†]	